EST II — MATH LEVEL 2 • UNIT 3.01

C Polynomial, Rational, Exponential & Logarithmic Functions

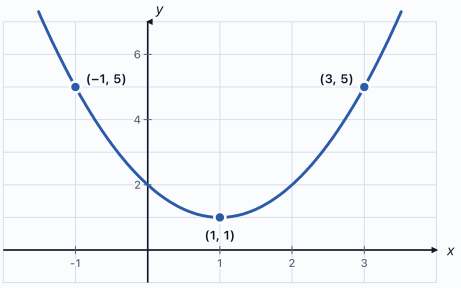
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Polynomial, Rational, Exponential & Logarithmic Functions Practice Exercises

1 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.



A 2

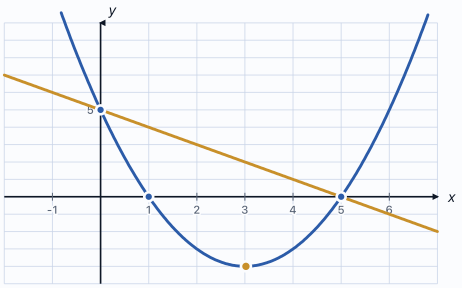
B 1

C 0

D -2

2 MCQ

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



A $\sqrt{25}$

B $\sqrt{16}$

C $\sqrt{36}$

D $\sqrt{49}$

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

A (2, 1)

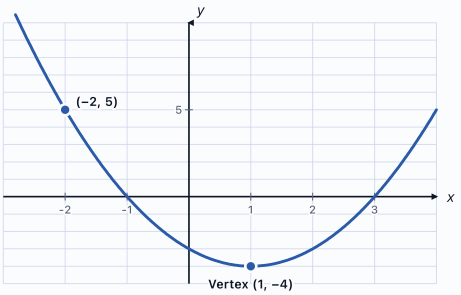
B (1, 0)

C (3, 0)

D (2, -1)

4 MCQ

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.



A 1

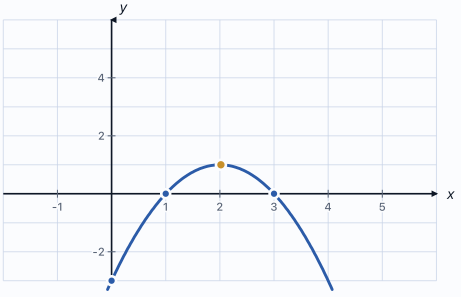
B -1

C 2

D -2

5 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.



A 3

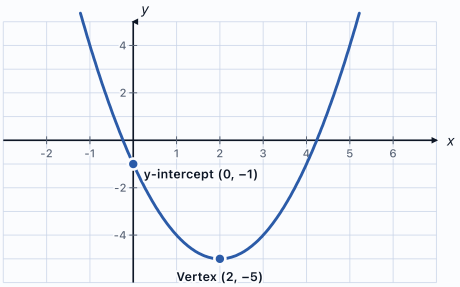
B 4

C 2

D 1

6 MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.



A -1

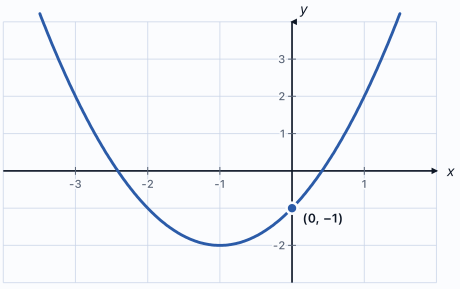
B 2

C 1

D -2

7 MCQ

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?



A -1

B 1

C 0

D 2

8 MCQ

If $f(x) = x^2$ and $g(x) = 2f(x - 1) + 8$, find $g(6)$.

A 80

B 58

C 56

D 60

9 MCQ

If $f(x) = x^2$ and $g(x) = 1f(x - 2) - 6$, find $g(-4)$.

A 31

B 10

C 30

D 28

10 MCQ

If $f(x) = x^2$ and $g(x) = 3f(x-6) - 5$, find $g(4)$.

A 43

B 7

C 5

D 10

11 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 2f(x-4) + 1$, find $g(3)$.

A 3

B 1

C 5

D 19

12 MCQ

Assuming that $f(x) = x^2$ and $g(x) = 3f(x-3) + 6$, find $g(3)$.

A 9

B 33

C 4

D 6

13 MCQ

Under the condition that $f(x) = x^2$ and $g(x) = 1f(x-6) + 7$, find $g(9)$.

A 17

B 16

C 88

D 14

14 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 1f(x-6) - 3$, find $g(2)$.

A 13

B 11

C 14

D 1

15 MCQ

If $f(x) = x^2$ and $g(x) = -1f(x-6) - 5$, find $g(4)$.

A -10

B -9

C -21

D -11

16 MCQ

Provided that $f(x) = x^2$ and $g(x) = 1f(x-1) + 4$, find $g(9)$.

A 68

B 69

C 85

D 66

17 MCQ

Assuming that $f(x) = x^2$ and $g(x) = 3f(x-4) - 2$, find $g(1)$.

A 28

B 25

C 1

D 23

18 MCQ

Under the condition that $f(x) = x^2$ and $g(x) = 1f(x-5) - 3$, find $g(5)$.

A -5

B -2

C 22

D -3

19 MCQ

If $f(x) = x^2$ and $g(x) = -1f(x-5) - 6$, find $g(-3)$.

A -15

B -72

C -70

D -71

20 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 3f(x-6) - 7$, find $g(-1)$.

A 138

B 140

C 143

D -4

21 MCQ

Find the domain of $f(x) = \sqrt{x+9}$.

A $x \leq -9$

B x>-9

C $x \in -9$

D $x \geq -9$

22 MCQ

Find the domain of $f(x) = \sqrt{x - 8}$.

A $x \leq 8$

B x>8

C $x \neq 8$

D $x \geq 8$

23 MCQ

Calculate the domain of $f(x) = \sqrt{x - 4}$.

A $x \leq 4$

B x>4

C $x \geq 4$

D $x \neq 4$

24 MCQ

Compute the domain of $f(x) = \sqrt{x - 0}$.

A $x \leq 0$

B x>0

C $x \neq 0$

D $x \geq 0$

25 MCQ

Calculate the domain of $f(x) = \sqrt{x + 1}$.

A $x \geq -1$

B $x \leq -1$

C x>-1

D $x \neq -1$